

Section 10.1 Worksheet: Floating Point and Partial Pivoting

Name: _____

Date: _____

1. Write each number in normalized floating-point form: -4382.7 , 0.0006712 , 95.03 .
2. Round each to four significant digits using round-half-to-even: 18.245 , 18.255 , 0.0099945 , 0.0099955 , -763.55 .
3. For exact $x = 0.0025$ and approximation $\hat{x} = 0.00248$, calculate absolute and relative error.
4. Explain catastrophic cancellation by evaluating $\sqrt{10001} - 100$ directly and after rationalization.
5. For
$$0.002x_1 + 2x_2 - 2x_3 = 6.002, \quad 3x_1 - 4x_2 - 4x_3 = -1, \quad x_1 + 4x_2 - 4x_3 = 13,$$
perform nonpivoted elimination with four-significant-digit rounding. Record every multiplier and matrix.
6. Repeat with partial pivoting. Explain each row swap and verify that every multiplier has magnitude at most one.
7. For both computed solutions, calculate $\|\hat{x} - x\|_\infty$ and $\|b - A\hat{x}\|_\infty$.
8. Compute or estimate $\kappa_2(A)$ for $A = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 1 & 1.0001 \end{bmatrix}$. Interpret.
9. Prove the right-hand-side perturbation bound.
10. Distinguish conditioning, forward stability, backward stability, and residual checking.

Workspace for elimination tables

Section 10.2 Worksheet: Jacobi and Gauss–Seidel Methods

Name: _____

Date: _____

For

$$10x_1 - x_2 + 2x_3 = 6, \quad -x_1 + 11x_2 - x_3 = 22, \quad 2x_1 - x_2 + 10x_3 = -10,$$

1. Derive all Jacobi update equations.
2. Derive all Gauss–Seidel update equations and clearly identify where new values replace old values.
3. Starting from $(0, 0, 0)^T$, complete the table.

k	Jacobi			Gauss–Seidel		
	x_1	x_2	x_3	x_1	x_2	x_3
0	0	0	0	0	0	0
1						
2						
3						

4. Form B_J and B_{GS} from $A = D + L + U$.
5. Compute the eigenvalues and spectral radius of each iteration matrix.
6. Prove convergence for Jacobi under strict diagonal dominance.
7. Prove convergence for Gauss–Seidel under strict diagonal dominance.
8. For $x + 4y = 5$, $3x + y = 4$, prove divergence in the displayed order and convergence after reordering.
9. Construct a stopping test using absolute tolerance, relative tolerance, residual, and maximum iteration count.
10. Derive the a posteriori contraction error bound.

Proof and iteration workspace

Section 10.3 Worksheet: Power Method

Name: _____

Date: _____

Let $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$.

1. Find the characteristic polynomial, eigenvalues, and bases for both eigenspaces.
2. Explain why 5 is the dominant eigenvalue.
3. Starting from $x^{(0)} = (1, 0)^T$, complete six scaled iterations using $\|\cdot\|_\infty$.

k	$x_1^{(k)}$	$x_2^{(k)}$	scale	$R_A(x^{(k)})$
1				
2				
3				
4				
5				
6				

4. Compute $\|Ax^{(k)} - R_A(x^{(k)})x^{(k)}\|_2$ for $k = 2, 4, 6$.
5. Prove convergence for a diagonalizable matrix with a unique dominant eigenvalue.
6. Show why the asymptotic convergence factor is $|\lambda_2/\lambda_1|$.
7. Compare expected iteration counts for ratios 0.2 and 0.9 to reduce error by 10^{-6} .
8. Give examples of failure caused by a missing dominant component and tied dominant magnitudes.
9. Explain sign alternation when the dominant eigenvalue is negative.

Iteration and proof workspace

Section 10.4 Worksheet: Numerical Applications

Name: _____

Date: _____

1. Starting from $S = \sum_i [y_i - (a_0 + a_1x_i + a_2x_i^2)]^2$, calculate all three partial derivatives and derive the quadratic normal equations.
2. For data $(0, 120), (1, 182), (2, 255), (3, 329), (4, 388), (5, 415)$, calculate every required sum.
3. Solve the resulting 3×3 normal equations by Gaussian elimination with partial pivoting. Show every row swap, multiplier, and back-substitution step.
4. Compute the fitted values, residuals, sum of squared error, and prediction at $x = 6$.
5. Derive the four scenic-overlook probability equations from the law of total probability and write the matrix system.
6. Starting from zero, perform eight Gauss–Seidel iterations in a table and compare with the exact solution.
7. For the Leslie matrix in the notes, perform ten power iterations, normalize to proportions, and estimate the growth factor with a Rayleigh quotient.
8. Explain why a stable distribution is an eigenvector and why the growth factor is its eigenvalue.
9. Compare numerical error, data uncertainty, and model-form uncertainty for all three applications.

Regression, probability, and population workspace